



Mandarin count estimation with 360-degree tree video and transformer-based deep learning

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ABSTRACT

In mandarin orange production, knowing the number of mandarins on a tree is crucial for performing appropriate thinning operations. This is necessary to prevent overbearing and to maintain the size and quality of individual mandarins. However, the dense foliage of mandarin trees often obscures many fruits. Before thinning, a tree can have several hundred to several thousand mandarins, making manual counting not only labor-intensive but also exceedingly difficult. This study proposes a novel technology for automatically estimating the number of small, unripe mandarins before thinning operations using 360-degree videos of the trees. A deep neural network model leveraging Transformer architecture is designed to fuse features from various viewing angles. State-of-the-art (SOTA) unsupervised pre-training techniques, along with a custom-designed loss function, are employed to tackle the challenge of limited training data and to enhance the model's capability to capture global contextual features from images taken at multiple angles. Experimental results demonstrate that the proposed method significantly outperforms existing methods.

1. Introduction

Predicting the number of fruits is useful for planning sales, allocating necessary labor, facilities, and materials [7]. Additionally, it is essential for deciding the extent of thinning required to prevent biennial bearing and maintain fruit size and quality [15]. Since manually counting fruits is time-consuming, it is not a feasible method for farmers. Consequently, the estimation of fruit numbers using computer vision and deep learning has been attracting considerable attention recently.

Several studies have been conducted on estimating the number of fruits. Wulfsohn et al. [27] sampled fruits from individual trees at the time of harvest to estimate total harvest volume. This study requires prior knowledge of the number and structure of branches for appropriate sampling. Additionally, miscounts can occur within the complex canopy. Anderson et al. [1] estimated the total harvest volume by sampling trees in the field at the time of mango harvest and examining the actual number of fruits on the sampled trees. This method incurs high manual labor costs. Additionally, there are studies predicting yield using Unmanned Aerial Vehicles (UAVs) [2,9,19], but UAVs may require special permits depending on the surrounding environment of the field and the laws and regulations of the country.

Dorj et al. [7] and Malik et al. [20] used a single RGB image of a mandarin tree along with traditional image processing algorithms to detect and count the fruits. This method is significantly affected by the weather and shooting environment when extracting features from RGB images. Kestur et al. [13] used a single RGB image of a mango tree and a deep neural network (DNN) model to detect and count the fruits. Chen et al. [5] used a single RGB image of a tree and a DNN to detect and count the fruits of apples and mandarins. Zhang et al. [30] used RGB videos and a DNN to detect and track fruits on a row of mandarin trees. These methods count the number of fruits visible from a particular shooting direction and do not estimate the total fruit count on the entire tree. Gao et al. [10] used RGB images taken from multiple viewpoints to count the visible apples on an entire tree. This method addresses the limitation of relying on a single image but still only accounts for visible fruits, neglecting those completely obscured by leaves. Given that mandarin trees are dense with foliage and many mandarins are hidden by leaves, it is crucial to account for those not visible on the surface; thus, merely detecting and tracking the visible ones is inadequate. Gongal et al. [11] estimated the number of fruits on apple trees by utilizing images captured from both the front and back of the tree. Koirala et al. [14] also estimated the number of fruits on mango trees using

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images taken from both the front and back of the tree, based on the number and density of the fruits visible in the images. These methods necessitate capturing the entire tree in a single image, a requirement not always practical in real-world settings due to spatial limitations. Although mango trees might have sparse foliage, making this approach more feasible, mandarin trees are typically more densely foliated, which can lead to a significant number of fruits remaining hidden from view when images are only taken from the front and back of the tree.

Recent studies in deep learning are advancing rapidly. In the field of computer vision, DNNs are now applied not only to images but also to video-related tasks. Current applications of DNNs in video include object detection and action recognition. Models such as ViViT [3] and TubeViT [21] have been proposed for these tasks. ViViT and TubeViT are based on the Vision Transformer (ViT) architecture (Dosovitskiy et al. [8]), which is known for its effective self-attention mechanism for capturing global context within images. ViT is designed for an input image size of 224×224 , leading to significantly increased computational and memory demands when processing video clips with a large number of frames. Adapting these models for video content introduces the challenge of excessive computational costs, making the direct application of ViViT or TubeViT to this study less feasible. However, employing a Transformer-based architecture like ViT could be beneficial for understanding the relationships between frames. The Video Swin Transformer, introduced by Liu et al. [18] based on the Swin Transformer by Liu et al. [17], incorporates the 3D Shifted Window-based Multi-Head Self-Attention (MSA) module. This module is predicated on the notion that video frames are sampled densely. Consequently, the model's ability to effectively capture features across frames may diminish with sparse sampling unless modifications are made.

Given the labor-intensive nature of manually counting the number of mandarins on trees to establish a ground truth dataset, determining how to effectively train the proposed model with a limited dataset represents one of the most significant challenges of this research. It is known that pre-training using a large dataset with similar features can be effective in solving this problem. However, when creating a custom DNN model tailored to the objectives of the research, there might not always be a large dataset available for pre-training. To adequately train our model even with only a small dataset, we propose a method inspired by recent unsupervised pre-training approaches, such as the Masked Auto Encoder [12], SimMIM [28], and Swin MAE [29]. These methods involve masking portions of images and reconstructing the pixel values for the masked areas using the information from the unmasked pixel regions. By enhancing the model's capability of referring to global context, this approach has proven to be effective for pre-training models like ViT and Swin Transformer. As shown with ConvNeXt V2 [26], employing training innovations such as skipping the feature extraction of the masked parts has proven effective even for CNN architectures. Additionally, as reported with Swin MAE, the Masked Auto Encoder method has proven to be effective even for small datasets. There are also studies like VideoMAE V2 [24] that extend the Masked Auto Encoder approach to videos to improve accuracy. However, as mentioned previously, direct usage of ViT models like ViViT is not feasible for the model proposed in this study. Therefore, it is necessary to customize the Masked Auto Encoder approach accordingly for our new model architecture.

In this study, we propose a novel system that leverages state-of-the-art deep learning techniques to estimate the number of small and unripe mandarins from RGB videos captured by mobile devices such as smartphones or tablets. This approach eliminates the need for specialized equipment, thereby offering a practical solution for farmers. Crucially, we address the challenge of occluded fruits by integrating multi-angle imagery, enabling the model to capture a more comprehensive global context of the tree canopy.

A key innovation of our work is the application of a Masked Autoencoder for unsupervised pre-training. By incorporating this state-of-the-art technique, along with a custom-designed loss function, we overcome the issue of limited training data and enhance the model's

ability to learn global contextual features from various viewing angles. We evaluated our proposed method using data collected from a major mandarin orange production region in Japan. Experimental results demonstrate that our system significantly outperforms existing approaches [5,14], highlighting both the effectiveness of the Masked Autoencoder-based pre-training and the practical value of our multi-angle estimation strategy. The main contributions of this paper can be summarized as follows:

- We propose a system that uses deep learning to estimate the number of small, unripe mandarins from RGB videos captured by mobile devices, offering a practical solution without the need for specialized equipment.
- We address occluded fruits by integrating multi-angle imagery, capturing a more comprehensive view of the tree canopy.
- We apply a Masked Autoencoder for unsupervised pre-training, enhancing the model's ability to learn global features from limited data.
- We evaluate the method using data from Japan's mandarin production region, showing superior performance compared to existing approaches.

2. Materials and methods

2.1. Data collection

Data collection on mandarins was conducted in Setoda Town, Onomichi City, Hiroshima Prefecture, Japan, in 2022 and 2023. A photograph of a mandarin orange field is shown in Fig. 1. The variety of mandarin is Early Satsuma Mandarin, and the age of the mandarin trees ranges from 15 to 40 years. The tablets used for filming were the iPad Pro 11-inch (4th generation) and the iPad Pro 12.9-inch (6th



Fig. 1. The image of mandarin farm.

generation). Mandarin trees were filmed using the camera function of a tablet device, ensuring that the entire tree fit within the frame as much as possible while circling around the tree for a 360-degree view. The filming criteria were as follows: shoot in vertical video format, include both the upper and lower parts of the tree within the frame as much as possible, and take measures to prevent out-of-focus shots and camera shake. Following this policy, the distance between the tree and the camera was approximately 1 m. However, the distance varied slightly depending on the space between the trees and the uneven footing in the field. These captured videos are in RGB format. The number of mandarins on each tree was manually counted to be used as the ground truth for training and validation. Data collection was conducted before the thinning operation, and the fruits on the mandarin trees were small and unripe.

Fig. 2 illustrates the distribution of the number of trees for which video data and the counted number of mandarins have been collected. The horizontal axis represents the range of the number of mandarins on a tree, while the vertical axis indicates the number of trees (videos) with mandarin counts falling within that range. Trees with an extremely low number of mandarins due to the alternate-bearing phenomenon were excluded, as they were not originally targeted for thinning. As a result, the range of mandarins counted on the collected trees was between 300 and 1200. The average number of mandarins was 685.19, with a standard deviation of 253.79, and the coefficient of variation was 0.37039. The collected data comprised videos of 89 trees with known numbers of mandarins and 73 trees with unknown numbers of mandarins.

2.2. Materials and methods overview of the proposed method

The outline of the proposed mandarin count estimation method is shown in Fig. 3. Once farmers have captured a 360-degree RGB video of a mandarin tree, a specified number of frames are sampled from the video (2.5). Mask images of the mandarins are created from each frame's RGB image (2.4), and then both the RGB images and the mask images are input into the estimation model to predict the number of mandarins (2.3). As presented in 2.1, the data on mandarin trees used in this study is limited. To address this issue, improvements to the loss function (2.6) and pre-training of the encoder using Masked Auto Encoder (2.7) are conducted during the training stage.

2.3. Proposed number estimation model

The structure of the proposed model is shown in Fig. 4. For the multiple frames obtained from the 360-degree video (see 2.5), feature extraction is performed for each frame using a Convolutional Neural Network (CNN), and the features from all frames are fed into a Temporal Swin Transformer to extract global features related to the number of mandarins. Finally, a Multi-Layer Perceptron (MLP) is used to estimate the number of mandarins utilizing the features obtained from the Temporal Swin Transformer.

2.3.1. Feature extraction for each frame

CNN takes an RGB image and the mask image of mandarins as input. The diagram of the CNN is shown in Fig. 5. To extract features from RGB images, the ConvNeXt V2 Block structure proposed by Woo et al. [26] was used. In this research, the RGB feature extractor was constructed based on ConvNeXt V2 nano. The channel and block settings of the stem layer and the first block were the same as those in Woo et al. [26]: the channel setting was 80, and the block setting of the first block was 2. The channel and block settings of the second block were changed from those in Woo et al. [26]: the channel setting was 160, and the block setting was 6. The third block and beyond were omitted. For feature extraction from the mask images of mandarins, a stem layer producing a 16-channel feature map and a simple convolutional layer with an activation function that also outputs a 16-channel feature map were utilized, comprising two layers as proposed by Woo et al. [26].

2.3.2. Feature extraction across frames

The Temporal Swin Transformer consolidates features extracted by CNNs from multiple frames and reconstructs the global features essential for estimating the number of mandarins on the entire tree. Fig. 6 shows the structure of the Temporal Swin Transformer. For each frame obtained from the CNNs, $P \times P$ patching and positional embedding were performed as in ViT (Dosovitskiy et al. [8]). The length of the input sequence for each frame was $length = \frac{WH}{P^2}$, where W and H are represented as the width and height of features extracted by CNNs and P is represent patch size, with token was D . Since this is a video task, there is a time axis that needs to be connected to the sequence axis. Video Swin Transformer proposed by Liu et al. [18] was designed to process 3D features through the Transformer architecture by incorporating both

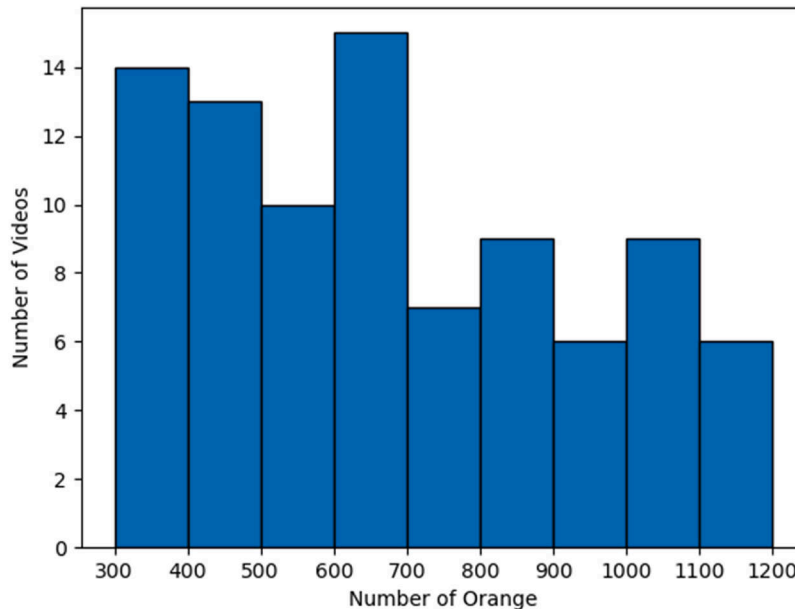


Fig. 2. . The number of collected videos(=trees) by the range of mandarin count.

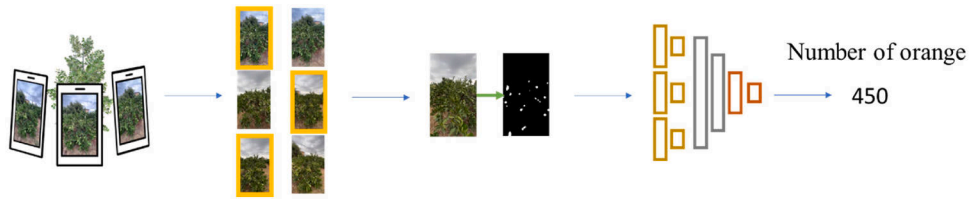


Fig. 3. Overview of the proposed method.

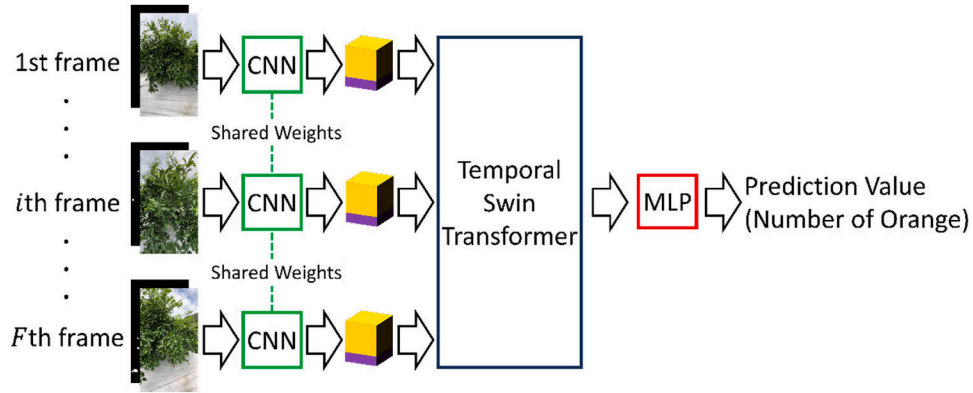


Fig. 4. Overview of the proposed number estimation model.

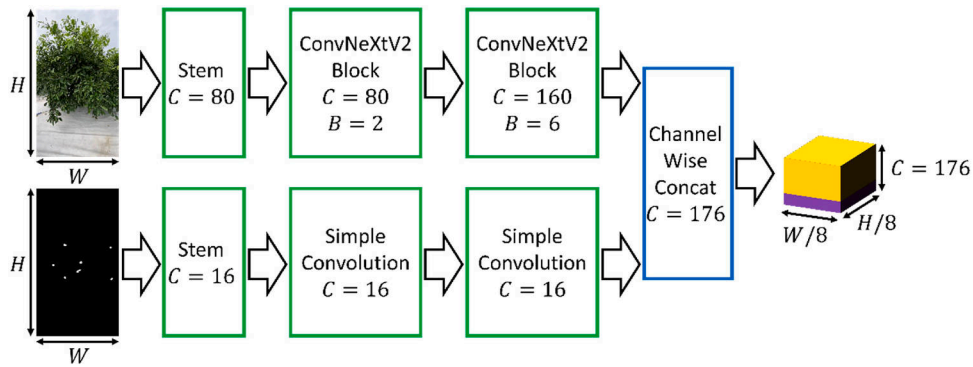


Fig. 5. The structure of CNN(C: Channel, B: Layer).

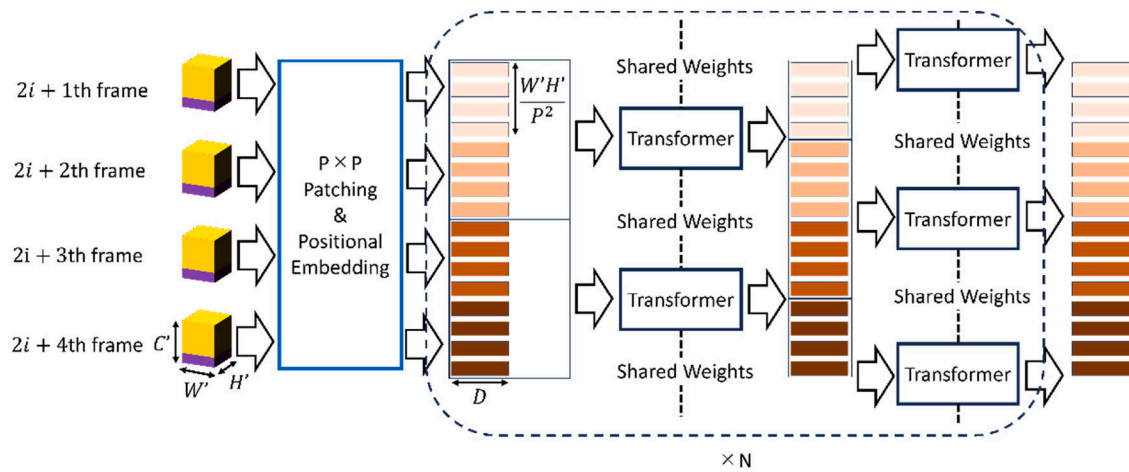


Fig. 6. The structure of temporal swin transformer (window size = 2).

spatial features and temporal features within windows. This dual approach aimed to leverage the intricate dynamics of spatial and temporal data for enhanced model understanding. However, the present study takes a different approach with its version of the Video Swin Transformer. Here, the model inputs 3D features exclusively through temporal windows, eschewing the spatial dimension. This methodological pivot is justified by the study's adoption of a sparse frame sampling technique, as opposed to a dense one, coupled with the observed low similarity between consecutive frames. In this study, tokens for each frame were connected to the sequence axis, so the length of the input sequence was $\frac{\text{window size} \times W \times H}{p^2}$ as shown in Fig. 6. These tokens were input to the Transformer to extract features between frames. To extract features between different adjacent frames from the first transformer, the tokens of each frame were reconnected to the sequence axis again by shifting the $\frac{\text{window size}}{2}$. These . These tokens were input to the second Transformer. Fig. 6 shows an example with a window size of 2, but the actual window size was set to 4. In addition, two layers of Temporal Swin Transformer were prepared ($N = 2$).

Furthermore, this study incorporates a structural enhancement by adopting the B2T (Bottom-to-Top) connection, as proposed by Takase et al. [23]. This connection balances stable convergence and the model's performance. According to the previous study by Takase et al. [23], the B2T connection has the ability to stabilize Transformer learning and improve prediction accuracy. Therefore, the B2T connection was chosen for this study.

2.3.3. Estimation of the number of mandarins from features

The MLP estimates the number of mandarins using the features obtained from the Temporal Swin Transformer. Fig. 7 shows the diagram of the proposed MLP, where the process is divided into two parts: one for feature extraction and aggregation for each patch obtained from the output of the Temporal Swin Transformer, and the other for feature extraction and aggregation within the same frame. The final fully connected layer (FC) estimates the number of mandarins from the features of all frames. LayerNorm was used before each FC layer for normalization purposes; while ReLU was used after each FC layer for activation. Since laying out all the features from all frames in one dimension from the start would require memory to store weights for the MLP and gradient values for backpropagation, which could decrease computational efficiency, our model aggregates the features using common weights until the number of features becomes sufficiently small. The output dimension of the features after passing through the FC and MaxPool layers was calculated by dividing the input dimension by n , as shown in Fig. 7. In the process of feature extraction and aggregation for each patch, the first FC layer was prepared by setting n as 2. In the first concat, the feature vectors were concatenated so that the features in each patch were aligned in the same frame. For the process of feature extraction and aggregation within the same frame, n was set as 2, 11, and 2 respectively for the MaxPool and two FC layers, as shown in Fig. 7. In the second concat, the feature vectors were concatenated so that the features from all frames were aligned into one feature vector. In the process of feature extraction and aggregation for all frames, n was changed to 2, 8, and 3 respectively for the MaxPool and two FC layers. Finally, the number of mandarins was obtained as the output.

2.4. Generation of mandarin mask videos

The generation of mandarin mask videos involves using an instance

segmentation model to detect the mandarins. For this purpose, the Mask2Former model [6] was utilized. The backbone of Mask2Former is Swin-S [17]. A model pre-trained on MSCOCO [16] was fine-tuned using images of mandarins and a custom annotation file created specifically for mandarin fruits. The results of the instance segmentation are shown in Fig. 8.

2.5. Selection of frames

The selection of multiple frames is key to successfully extracting informative features required for mandarin count estimation. Mandarins can be occluded by leaves in images captured from certain angles. By using multiple frames taken from different viewing angles, it is possible to reveal mandarins that are otherwise hidden. To obtain an informative set of frames, it is essential to avoid selecting frames captured from similar angles.

This study does not utilize fruit tracking, as proposed by Zhang et al. [30], or structure-from-motion methods. The primary reason is that both approaches present significant limitations for our specific use case. Fruit tracking and structure-from-motion methods face challenges when dealing with occlusions, particularly in videos captured while rotating around an object. In this study, mandarins are often occluded by leaves, and the complex shape of the tree canopy makes it extremely difficult to maintain consistent tracking of the fruits.

To address these challenges, we employed a different approach: selecting 2D frames using two distinct methods—one based on the required number of frames and the other on the accumulated differences between frames. This approach avoids the complexity associated with 3D tracking or motion-based methods, offering a simpler yet effective way to ensure adequate visual coverage of the tree, even in the presence of occlusions and a dense canopy. The proposed frame selection method provides a robust solution while circumventing the limitations inherent in fruit tracking and structure-from-motion techniques.

Fig. 9 shows an example of frame selection based on the required number of frames, while Fig. 10 shows an example of frame selection based on accumulated differences between frames. Both Fig. 9 and Fig. 10 are derived from frames selected from the same video, demonstrating two different approaches to selecting frames from identical input data.

2.6. Based on the required number of frames

In this method, the i^{th} representative frame is selected based on the value calculated using the following formula:

$$\text{frame}_i = \left\lfloor \frac{N}{F} * i \right\rfloor + 1 \quad i = 0, 1, \dots, F - 1 \quad (1)$$

where F is the required number of frames to be input into the deep learning model, N is the total number of frames in the video.

This is the simplest method and requires the least amount of time to extract frames. However, it does not account for variations in camera movement speed during video recording, assuming that the camera moves at a constant speed along the circumference throughout the recording. Consequently, this method may result in sampling more similar frames from angles where the camera moves slowly, while insufficient frames may be captured from angles where the camera moves quickly.

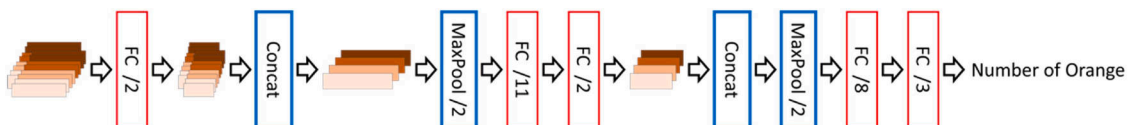


Fig. 7. The structure of MLP.



Fig. 8. Detection results of mandarins and their mask images. Left images are raw RGB images. Center images are the result of mandarins detection. Right images are mandarins mask images.

2.6.1. Based on accumulated difference between frames

The basic idea of this method is that an informative frame set should consist of frames that exhibit significant differences from each other. To select such frames, we define *Diff* as a metric to measure the degree of difference between two frames. We utilize Structural Similarity Index (SSIM) [25] a metric for evaluating the similarity of image structures, to calculate the degree of difference *Diff*. Since SSIM values range from 0 (least similar) to 1 (most similar), *Diff* can be computed as:

$$Diff = 1 - SSIM \quad (2)$$

As the camera gradually moves, frames that are immediately adjacent typically exhibit a high degree of similarity. Therefore, we accumulate the *Diff* values between adjacent frames and select the frames

when the accumulated *Diff* reaches or exceeds a certain threshold. Denoting the number of frames to be selected as F and the total accumulated *Diff* values of all adjacent frame pairs as SUM , then the threshold T_i for selecting i^{th} frame can be computed as:

$$T_i = SUM * i / F \quad i = 0, 1, \dots, F - 1 \quad (3)$$

Below is the pseudo-code of the algorithm for selecting frames based on the accumulated *Diff* values. It consists of two steps. The first step builds a list of accumulated *Diff* values, with each element representing the accumulated *Diff* for the corresponding frame. The second step selects representative frames based on this accumulated *Diff* list.

Algorithm 1



Fig. 8. (continued).



Fig. 9. Examples of frame selection based on the required number of frames.

2.7. Improvement of loss function for preventing overfitting

Typically, regression problems use L1Loss or L2Loss. These loss functions aim to force the model's predicted values to match the ground truth values exactly. In datasets with a wide range of ground truth values and limited dataset size, it is conceivable that the model could overfit, leading to decreased accuracy on unseen data. Therefore, this study

proposes the following loss function.

$$\text{WeakL1Loss} = \text{Max}(\text{L1Loss} - \alpha * \text{Ground Truth}, 0) \quad (4)$$

This loss function aims to stop training when the predicted value falls within an error tolerance range of $\pm \alpha * \text{ground truth}$. This approach allows for a tolerance in the prediction error, helping to prevent the model from overfitting. The proposed loss function selectively focuses on data



Fig. 10. Examples of frame selection based on accumulated difference between frames.

Algorithm 1

Build frame list by accumulated diff.

```

1. def get_frame_list(frames: List[frame], F = 12: Number)
2. # first step
3. number_of_frames = len(frames)
4. accumulated_diff_list = [0]
5. for i in range(number_of_frames - 1):
6. diff = 1 - SSIM(frames[i], frames[i+1])
7. accumulated_diff_list.append(accumulated_diff_list.last_element() + diff)
8. # second step
9. sum = accumulated_diff_list.last_element()
10. frame_list = []
11. for j in range(F):
12. target_accumulate_sum = sum * j / F
13. target_frame_idx = get_first_idx_whose_element_has_a_value_not_less_than_val(accumulated_diff_list, target_accumulate_sum)
14. frame_list.append(frames[target_frame_idx])
15. return frame_list

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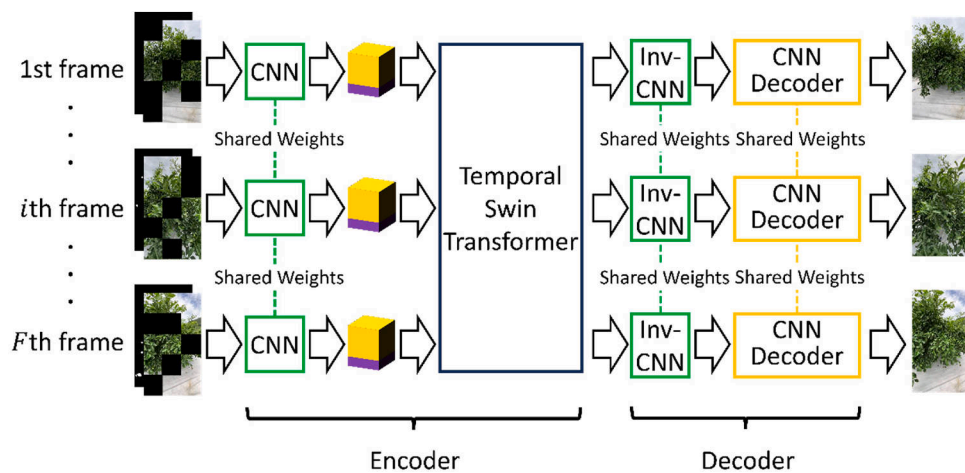


Fig. 11. The model for pre-training.

with larger errors, enabling for targeted improvement in challenging cases while avoiding overfitting to data with already small errors. This approach differs from regularization and early stopping, which are applied globally across all data. Instead, our method emphasizes continuous learning for data points that require the most adjustment, ultimately improving the model's robustness without unnecessary overfitting.

2.8. Application of masked auto encoder

The methods proposed by Xu et al. [29] and Wang et al. [24] apply the Masked Auto Encoder technique [12] to efficiently train Transformers with a small dataset to achieve improved accuracy. In this study, we propose using the Masked Auto Encoder technique for pre-training the CNN and Temporal Swin Transformer of the proposed model. The model used for the Masked Auto Encoder is shown in Fig. 11. This model simply replaces the MLP part of the proposed model with a Decoder using Inv-CNN and FC, as proposed by Woo et al. [26]. Although the frames are shot from different perspectives, some may capture overlapping areas. The aim of this Masked Auto Encoder is to use the Temporal Swin Transformer to reconstruct images, enabling it to extract contextual features across frames. The input consists of masked images, and the output is reconstructed images. However, masking is only performed on the RGB images, not on the mask images of the mandarins. This enables the model to capture features across frames using the Temporal Swin Transformer, as well as features between the RGB and mask images. Additionally, since this Masked Auto Encoder does not require the number of mandarins as ground truth, it can utilize data where the number of mandarins is unknown. In this study, as explained in 3.1, videos of trees whose number of mandarins is unknown were used for pre-training.

3. Experiments and results

3.1. Creation of training and validation datasets

Although the dataset is relatively small, with 89 trees, k-fold cross-validation was used to ensure robust validation and effectively handle the limited amount of labeled data. This method was chosen to make the most out of the available dataset and to ensure that the proposed model can generalize well. Given the imbalance in the number of videos per number of mandarins, as shown in Fig. 2, it was deemed appropriate to create training and validation datasets while maintaining this imbalance. First, groups of G videos were created in ascending order of the number of mandarins. Then, K -fold division was performed within each video group. In the current implementation, $G = 12$ and $K = 5$ were chosen. Table 1 shows the number of sub-datasets split by stratified K -fold cross-validation.

To increase the amount of training data, multiple sets of F frames were sampled from the same video by shifting the starting frame when selecting F frames from the candidate list. This approach allows for generating more diverse data from the same video by capturing different perspectives, thus ensuring better model generalization. The validation and test datasets, however, were sampled only once to maintain consistency in evaluation. This data augmentation strategy helps effectively increase the training set size without requiring additional data collection efforts, making it particularly useful in scenarios with limited available videos. The added variability from sampling different sets of frames within the same video contributes to improving the robustness of the model by exposing it to different viewpoints and conditions.

Table 1
Number of trees in groups split by stratified K-fold cross-validation.

K	1	2	3	4	5
Number of videos	18	18	19	16	18

However, it is important to note that over-reliance on a single video source may lead to overfitting to specific conditions. Therefore, using a diverse range of videos is advisable to improve generalization. In our research, due to the limited number of available videos, this method was employed to ensure sufficient training of the deep learning model. This technique is likely applicable to other video-related tasks as well, making it a versatile approach for expanding training datasets in various video analysis scenarios.

For the pre-training dataset used with the Masked Auto Encoder, data from the 73 videos of trees with unknown numbers of mandarins were also included in each training set after stratified K-fold splitting. The Masked Auto Encoder-based pretraining uses an inpainting task rather than the mandarin number counting task to enhance the feature extraction backbone's ability to capture contextual information across images taken from different view angles. This pretraining method, which does not require the ground truth of mandarin numbers, provides an effective way to address the challenge of obtaining sufficient training data with known mandarin counts. To increase the training data, multiple samplings were conducted from the same video, even for videos of trees with unknown numbers of mandarins.

3.2. Experimental method

3.2.1. Searching for the optimal number of frames

We conducted experiments to determine the appropriate number of frames F , as described in 2.5 for selecting the multiple frames into be input to the CNN. Considering the RGB videos filmed around the circumference of the tree and taking into account overlaps between the sampled frames, we experimented with three conditions for $F \in \{12, 24, 36\}$. The experiments were conducted with the settings shown in Table 2, training on a dataset of trees with known numbers of mandarins, using a batch size of 6. The learning rate decreased from the original rate to the minimum rate using Cosine Annealing. Data augmentation with Random Brightness Contrast was applied with the same parameters to the F frames of RGB images in one set of training data. Additionally, Random Resized Crop, Horizontal Flip, and Rotation were applied with the same parameters to the F frames of both RGB images and mandarin mask images in one set of training data.

3.2.2. Pre-training with masked auto encoder

Pre-training with the Masked Auto Encoder was conducted with the settings shown in Table 3. Training was performed with a batch size of 6. A warmup was used to gradually increase the learning rate from the minimum learning rate during the first 15 epochs, after which the learning rate was decreased to the minimum learning rate using Cosine Annealing. Data augmentation with Random Brightness Contrast was

Table 2
Experimental settings for searching the optimal number of frames.

Config	Settings
	From scratch
optimizer	AdamW
base learning rate	$1e-4$
min learning rate	$1e-6$
weight decay (CNN and Temporal Swin Transformer)	0.05
optimizer momentum	$\beta_1, \beta_2 = 0.9, 0.95$
base batch	6
learning rate schedule	cosine decay
warmup epoch	3
training epoch	30
Random Brightness Contrast	$p, \text{brightness, contrast} = 0.75, \pm 0.2, \pm 0.2$
Random Resized Crop	$p, \text{scale, ratio} = 0.7, [0.8, 1.0], [0.833, 1.2]$
Horizontal Flip	$p = 0.5$
Rotate	$p, \text{degree} = 0.5, \pm 7$

Table 3
Experimental settings for pre-training with masked auto encoder.

Config	Settings
optimizer	AdamW
base learning rate	1e-5
min learning rate	1e-8
weight decay	0.05
optimizer momentum	$\beta_1, \beta_2 = 0.9, 0.95$
base batch	6
learning rate schedule	cosine decay
warm up epoch	15
training epoch	150
Random Brightness Contrast	$p, \text{brightness}, \text{contrast} = 0.75, \pm 0.2, \pm 0.2$
Random Resized Crop	$p, \text{scale}, \text{ratio} = 0.7, [0.8, 1.0], [0.833, 1.2]$
Horizontal Flip	$p = 0.5$

applied to the 12 frames of RGB images for each batch of training data using the same parameters. Furthermore, Random Resized Crop and Horizontal Flip were applied to both the 12 frames of RGB images and the mandarin mask images for each batch of training data using the same parameters.

3.2.3. Training for mandarin count estimation

The training for mandarin count estimation was conducted with the settings described in Table 4. Training was performed on the dataset of videos from trees with known mandarin numbers, using a batch size of 6. The learning rate was decreased from the original learning rate to the minimum learning rate using Cosine Annealing. When fine-tuning from the Masked Auto Encoder, weight updates for the CNN part were stopped, and the learning rate for weight updates of the Temporal Swin Transformer was reduced using layer-wise learning rate decay as proposed by Bao et al. [4]. Data augmentation with Random Brightness Contrast was applied to the 12 frames of RGB images for each batch of training data using the same parameters. Furthermore, Random Resized Crop, Horizontal Flip, and Rotate were applied to both the 12 frames of RGB images and the mandarin mask images for each batch of training data using the same parameters.

3.2.4. Evaluation metric

In our experiment, we used the Mean Absolute Error (*MAE*) and Mean Absolute Percentage Error (*MAPE*) as evaluation metrics for the model. *MAE* is calculated by taking the average of the absolute differences between the predicted number of mandarins and the actual counts, with its unit being the number of mandarins in this study. *MAPE* is calculated by dividing the error of each prediction by the actual value, taking the absolute value, and then averaging these values for all the test

Table 4
Experimental settings for mandarin count estimation.

Config	Settings	
	From scratch	Finetuning
optimizer	AdamW	
base learning rate	1e-4	
min learning rate	1e-6	
weight decay (CNN and Temporal Swin Transformer)	0.05	
optimizer momentum	$\beta_1, \beta_2 = 0.9, 0.95$	
base batch	6	
learning rate schedule	cosine decay	
warmup epoch	3	
training epoch	30	
CNN frozen weights	No	Yes
layer-wise lr decay (only temporal swin transformer)	No	0.85
Random Brightness Contrast	$p, \text{brightness}, \text{contrast} = 0.75, \pm 0.2, \pm 0.2$	
Random Resized Crop	$p, \text{scale}, \text{ratio} = 0.7, [0.8, 1.0], [0.833, 1.2]$	
Horizontal Flip	$p = 0.5$	
Rotate	$p, \text{degree} = 0.5, \pm 7$	

data, with the unit being a percentage. The definitions of *MAE* and *MAPE* are given by Eqs. (5) and (6), respectively:

$$MAE = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i| \quad (5)$$

$$MAPE = \frac{100}{N} \sum_{i=1}^N \left| \frac{\hat{y}_i - y_i}{y_i} \right| \quad (6)$$

Where, N is the dataset size, y_i is the actual value, $\hat{y}_i$ is the predicted number. Moreover, the *MAE* and *MAPE* shown in 3.3 are the averages of the *MAE* and *MAPE* calculated from each validation dataset split by k -fold cross-validation.

3.3. Experimental results

3.3.1. Optimal number of frames

Trained with the parameter settings described in 2.2, the results of the mandarin count estimation using 3 different numbers of frames are shown in Table 5. The best results for both *MAE* and *MAPE* are achieved when $F = 12$. Therefore, we adopted 12 frames for the experiments shown in the remaining part of this section.

3.3.2. Results of mandarin count estimation

The detailed results are shown in Table 6. The best average *MAE* and *MAPE* values were achieved with the model pre-trained with Masked Auto Encoder, using the frame number-based selection, and trained with WeakL1Loss. The results of the two-tailed t -test analysis showed no significant differences between the proposed methods at the 95 % confidence level for both *MAE* and *MAPE*.

Table 7 presents a comparison of the proposed method with two existing approaches: the single-image-based method by [5] and the dual-image method, which employs images captured from both the front and back of the trees, by [14]. In the reference [14] paper, the final mango count is derived by multiplying the mango count detected from two images by an occlusion factor. This factor is computed as the average ratio of the visible mango count visible in the two images over the ground truth count. In our study, both of these existing methods were re-implemented using the same mandarin dataset that we collected, ensuring a consistent basis for comparison. The results of the analysis using the two-tailed t -test showed that the proposed method achieved significantly higher accuracy than both of the existing methods at the 95 % confidence level in both *MAE* and *MAPE*, underscoring its effectiveness in estimating mandarin counts.

4. Discussion

Regarding the number of frames used, the best results were achieved when using 12 frames ($F = 12$). This may be attributed to the fact that, in deep learning models, an increase in the number of input frames escalates the number of parameters requiring optimization. Consequently, with a limited dataset, the complexity of successfully training the model increases.

The mandarin count, as determined by human ground truth, was divided into intervals of 100 for further analysis. *MAE* and *MAPE* values were calculated for each interval to assess model performance. In the 300–500 and 700–1200 intervals, the model pre-trained with a Masked Auto Encoder demonstrated significant improvements in both *MAE* and

Table 5
Results of mandarin count estimation when using different number of frames.

Input frames	MAE	MAPE
12	208.70	28.63
24	224.17	31.19
36	227.81	31.26

Table 6

Average MAE and MAPE for trees bearing the number of mandarins.

	Without Masked Auto Encoder				With Masked Auto Encoder			
	L1Loss		WeakL1Loss		L1Loss		WeakL1Loss	
	MAE	MAPE	MAE	MAPE	MAE	MAPE	MAE	MAPE
Frame Number Based	208.70	28.63	223.88	30.81	194.02	27.08	189.56	26.12
Cumulative Sum Based on SSIM	216.03	30.90	238.97	33.19	195.99	27.86	190.38	26.74

Table 7

Comparison with Existing Methods.

	MAE	MAPE
1 image method [5]	632.28	104.18
2 images method [14]	251.88	43.70
Proposed Method	189.56	26.12

MAPE compared to the non-pre-trained model, although these improvements were not statistically significant based on the Kruskal-Wallis test. In contrast, in the 500–700 interval, the non-pre-trained model outperformed the pre-trained model, showing lower MAE and MAPE values. However, the Kruskal-Wallis test again indicated no statistically significant differences in any of the intervals. These findings suggest that while pre-training may offer performance benefits in certain ranges, it does not consistently outperform the model without pre-training across all human-verified ground truth intervals.

The application of Masked Auto Encoder enables CNNs and the Temporal Swin Transformer to better extract features from multiple frames. Specifically, the improved MAE and MAPE values across many ranges, as shown in Figs. 12 and 13, indicate that the use of the Masked Auto Encoder is effective. Additionally, these results show that the Temporal Swin Transformer, which typically requires a large amount of data for training, is capable of extracting features even with a small amount of training data.

As shown in Figs. 12 and 13, the data for trees with a high number of mandarins is limited. While previous studies have focused only on the fruits visible on the surface of the tree, this study estimates the total number of fruits, including those that are hidden inside the tree. The accuracy of the data for trees with a high fruit count is lower, indicating that estimating the total number of fruits, including those hidden inside the tree, is challenging with the method proposed here.

Fig. 14 shows the Absolute Error for each tree, providing a deeper understanding of the error distribution across different tree instances. We observed that trees with a higher number of mandarins tend to exhibit higher Absolute Error values. This suggests that as the density of

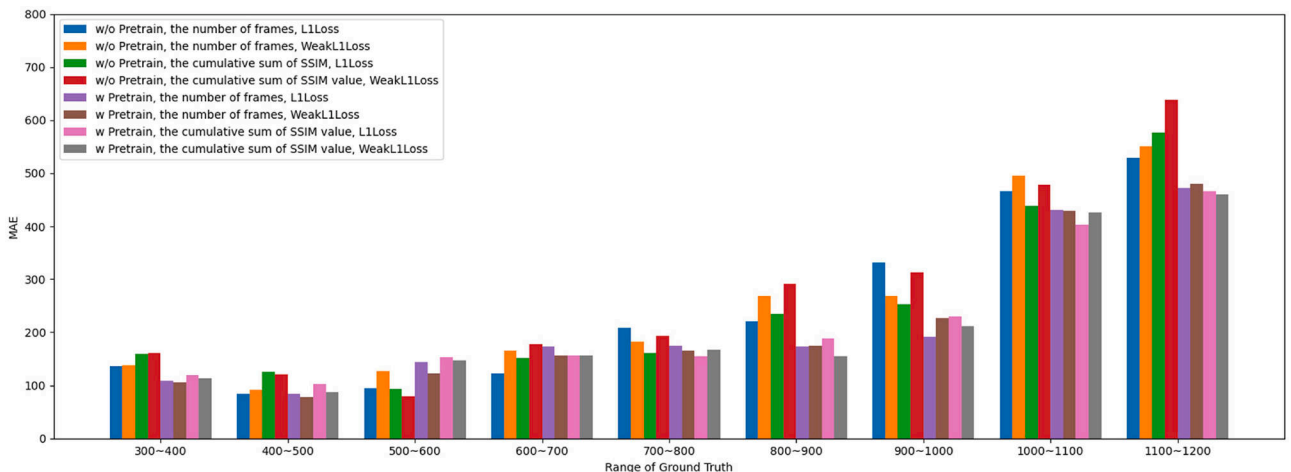
fruits increases, the complexity of accurately estimating their count also grows, likely due to increased occlusion and overlapping fruits. These conditions make it more challenging for the model to distinguish individual fruits, leading to greater prediction errors.

Table 8 and Fig. 15 present three representative examples. One each from the 300–400 range, 700–800 range, and 1100–1200 range. These examples corroborate the broader trends noted above. Tree A, with around 360 mandarins, shows a moderate Absolute Error (59), partly due to fewer occlusions compared to denser canopies. Tree B, with approximately 720 mandarins, also maintains a relatively moderate error (48.6), suggesting that the model can still handle a medium-level fruit density with reasonable accuracy. In contrast, Tree C, which has >1100 mandarins, exhibits a substantially higher Absolute Error (310). This result highlights the increasing complexity of accurate fruit counting when the tree's canopy is densely populated, as heavy occlusions and overlapping fruits significantly challenge the model's ability to detect individual mandarins.

On the other hand, trees with a moderate or lower number of mandarins generally show lower Absolute Error values, indicating that our model performs better in less complex scenarios where fruits are more distinct and less crowded. The observed trend aligns with our expectations, as the visual complexity of densely packed fruit clusters presents significant challenges for accurate prediction. These factors have increased the variance of the Absolute Error, which likely contributed to the lack of statistical significance in the two-tailed *t*-test.

This analysis highlights the need for more sophisticated approaches to improve the model's performance in high-density scenarios, such as incorporating additional image viewpoints or enhancing the model's ability to handle occlusions. These improvements could help reduce variance in the Absolute Percentage Error and increase overall prediction accuracy, particularly for trees with large numbers of mandarins.

We would also like to highlight that the complexity of accurately estimating fruit count is influenced by several factors such as canopy shape and density, which cannot be easily captured by traditional dimension measurements. Our system's reliance on image data allows it to directly consider these visual characteristics, which are essential for

**Fig. 12.** The MAE values for each interval of mandarins numbers.

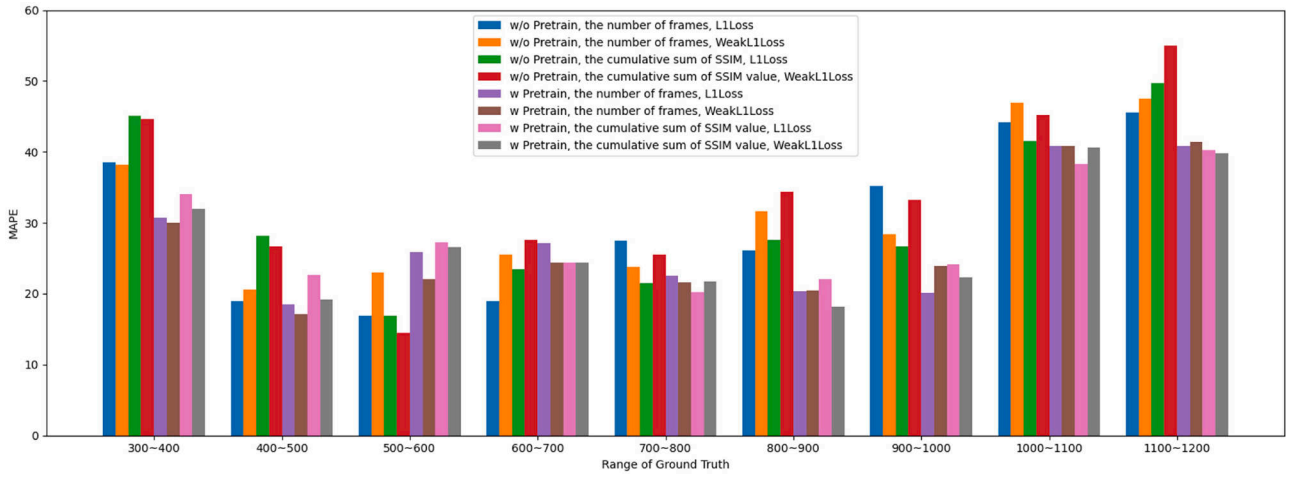


Fig. 13. The MAPE values for each interval of mandarins numbers.

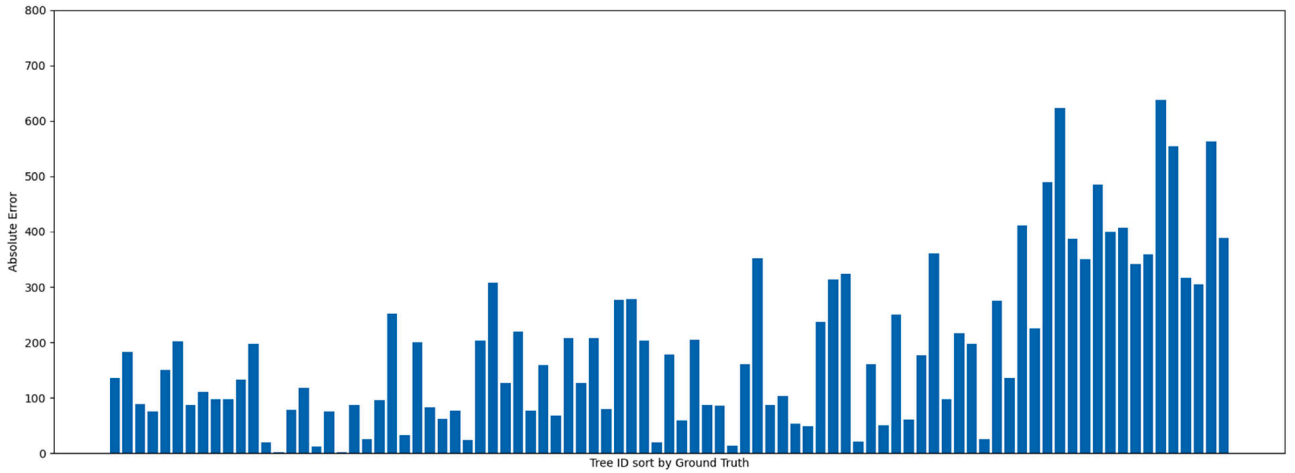


Fig. 14. The Absolute Error for each orange tree ordered by ground truth number of orange in tree under condition With Masked Auto Encoder, WeakL1Loss, Cumulative Sum Based on SSIM.

Table 8

Representative examples of actual and predicted counts for individual trees (Each Corresponding to a Different Ground Truth Range).

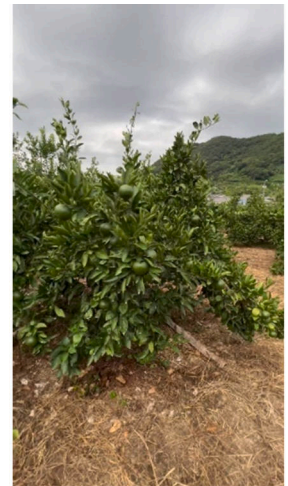
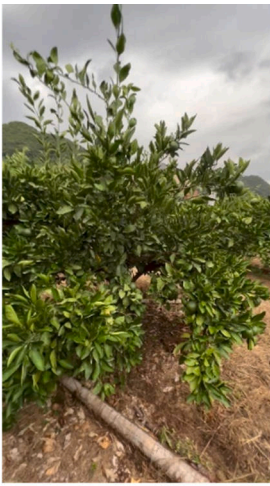
	Actual Counts	Predicted Counts	Absolute Error
Tree A	364	423.134	59.134
Tree B	724	675.384	48.616
Tree C	1175	864.808	310.192

improving the accuracy of fruit count estimation. By accounting for the tree's visual appearance, our system can adapt to the variability in canopy structure and fruit distribution that is often overlooked by relying solely on physical dimensions.

Regarding the effectiveness of the two frame selection methods, experimental results indicate that, overall, the frame-number-based sampling approach performed slightly better than the accumulative difference-based approach. The latter was proposed under the assumption that non-constant camera speed might lead to the selection of more frames with minimal changes during slow movements. The goal of selecting frames with significant accumulated differences was to ensure an informative frame set. However, this approach may not be as effective when the camera moves at a constant speed. Additionally, vertical camera movements can introduce larger differences between adjacent frames, further challenging the effectiveness of the accumulative

difference method. Table 9 presents the accuracy of both methods for three trees in the test data. Tree D achieved better results with the accumulative difference method, Tree F favored the frame-number method, and Tree E yielded almost identical results for both approaches. Fig. 16, showcasing camera trajectories generated with the COLMAP [22] technique, illustrates that Tree D, benefiting from the accumulative difference method, features intricate curves and slower movements at those points. Tree E's trajectory is smoother, yet includes some curves. Conversely, Tree F, where the frame-number method proved more effective, exhibits a less complex trajectory with generally faster movement. The correlation between trajectory differences and method efficacy suggests that the accumulative difference method is suitable for non-uniform camera movements, aligning with our proposal's intent. For future work, we plan to develop a technique to build the optimal frame set based on camera parameter values.

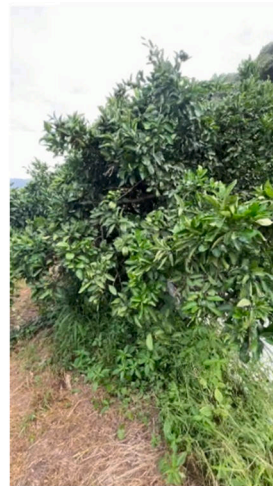
While fruit tracking methods aim to follow individual fruits across multiple frames to improve accuracy, this becomes challenging in situations with frequent occlusions, such as when fruits are completely hidden behind leaves or branches and are not visible from any viewpoint. Since there are correlations between the number of occluded fruits and the number of visible fruits, along with the density of leaves, we expect that by employing an end-to-end learning approach, the model can learn the relationship between the actual fruit count and the visual features of the tree, enabling it to predict the actual number with



(a) RGB images of Tree A in four directions



(b) RGB images of Tree B in four directions



(c) RGB images of Tree C in four directions

Fig. 15. RGB images of trees in four directions in [Table 8](#).

Table 9

Comparison of frame selection methods for individual trees.

	Absolute Error	
	Frame Number Based	Cumulative Sum Based on SSIM
Tree D	198.68	60.11
Tree E	86.96	87.05
Tree F	90.69	204.77

reasonable accuracy.

Similarly, 3-D localization methods provide precise spatial information, enabling an accurate count of fruits in three dimensions. However, they cannot detect fruits that are completely occluded from all viewpoints. Additionally, these methods require specialized hardware, are computationally intensive, and may be difficult to implement in environments where trees are densely packed. In contrast, our method focuses on 2-D frame analysis, making it simpler, less resource-intensive, and more practical for use with standard image data. This approach offers a more feasible solution for most commercial orchard settings.

However, it is important to acknowledge the limitations of our approach. The results can vary depending on the frames selected, which may lead to inconsistencies in measurements. Additionally, significant scale changes between frames can complicate accurate estimation, especially when fruits are occluded or vary in size. These challenges are reflected in the Mean Absolute Error (MAE) values, indicating that there is still room for improvement in the consistency of our method. Despite these limitations, our frame selection approach remains a practical alternative when more sophisticated tracking or 3-D techniques are not feasible due to hardware, environmental, or computational constraints.

Further enhancements are expected with the expansion of the dataset. The number of mandarins on a tree is influenced by numerous factors, including the current year's weather conditions, the previous year's yield, soil properties, and nutritional status. So far, we have validated the accuracy using data collected over two years from several distinct farms within the same region. However, since the estimation relies solely on features observed in the captured images, we anticipate that

the model could be applicable to a wide range of trees under various conditions. Nevertheless, the model's generalizability needs to be confirmed with data from different regions over multiple years.

In this study, we photographed the perimeter of the trees, which required sufficient space between them. The commercial farm where we collected data had enough space to accommodate this, as is typical for standard agricultural activities like thinning, fertilizing, and harvesting. However, we acknowledge that not all farms may have the same amount of space available. This requirement represents one of the limitations of our approach, as adequate space is essential for effective data collection.

5. Conclusions

In this study, we proposed a system for estimating the number of small and unripe fruits on mandarin trees from RGB videos using SOTA deep learning technologies. The proposed solution leverages mobile sensing devices and AI technology to assist farmers in estimating the number of fruits on mandarin trees. This task, though highly labor-intensive, is essential for maintaining the quality of mandarins. By utilizing multiple images taken from different viewing angles, the proposed method successfully predicts the number of mandarins, including those obscured by leaves from specific angles, with an average error below 27 % using the *MAPE* evaluation. This significantly outperforms existing methods that rely on a single image or just two images.

The proposed model can be easily adapted to estimate the number of other fruits, particularly those that may be occluded by dense foliage—a task that existing methods have struggled to achieve.

Data availability

The data that support the findings of this study are available from the corresponding author, Mao, upon reasonable request.

Ethics statement

Not applicable: This manuscript does not include human or animal

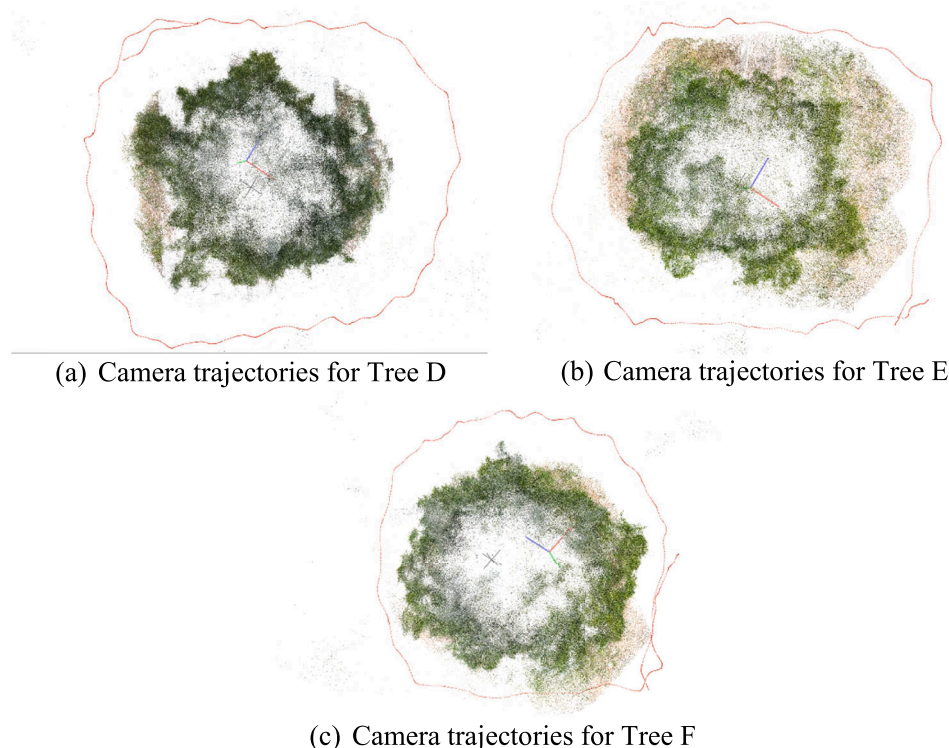


Fig. 16. Camera trajectories for the trees in Table 9.

research.

CRedit authorship contribution statement

Daisuke Isobe: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Prawit Buayai:** Writing – review & editing, Methodology, Investigation. **Xiaoyang Mao:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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